

Strategic Taxonomy in Heterogeneous Landscapes: Dynamics of Topological Collapse and Convexity Detection

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Abstract

Classical game theory often assumes static payoff matrices. This paper proposes a dynamic framework for "Heterogeneous Landscapes," where the topological class of a game (Zero-Sum, Negative-Sum, Positive-Sum) is mutable based on agent behavior and trust decay. We formalize the *Trust Parameter* $\tau(t)$ as a state variable governed by differential dynamics and derive the critical threshold τ_c , below which Positive-Sum systems irreversibly collapse into Negative-Sum "Pyrrhic Traps." Furthermore, we formalize the "Mustard Seed" phenomenon as a Signal Extraction Problem, distinguishing between observers optimizing for current state variance (Type S) and those detecting latent convexity (Type V). We present the *Loyalty Filter Algorithm*, provide bounds on recovery versus decay, and validate the topological collapse via agent-based simulations.

Nomenclature

Symbol	Definition
Ω	The universal set of strategic interactions
$\Psi(a)$	Aggregate Potential (Total Utility) of action profile a
$\tau(t)$	Dynamic Trust Parameter, $\tau \in [0, 1]$
τ_c	Critical Trust Threshold (Bifurcation Point)
$\lambda(t)$	System Friction / Entropy / Transaction Costs
$\mathcal{P}(t)$	Latent Potential (Convexity of success function)
$S(t)$	Observable Success Signal
κ	Coupling Coefficient in synergistic interactions
W	Work/Energy invested in trust recovery
$\Omega_I, \Omega_{II}, \Omega_{III}$	Topological Classes: Zero-Sum, Negative-Sum, Positive-Sum

1 Introduction

Classical Game Theory, established by Von Neumann and Morgenstern [1], typically isolates strategic interactions into static categories. However, agents in real-world systems navigate a heterogeneous landscape where the topology of the game itself is unstable. A cooperative partnership (Positive-Sum) can degenerate into a lawsuit (Negative-Sum) due to a single defection event.

The central thesis of this paper is that the gravest errors in strategic decision-making arise from a *Categorical Error*: misidentifying the topological class of the game. We categorize the strategic manifold Ω into three domains based on the gradient of the Aggregate Potential Ψ :

1. **Class I (Zero-Sum, $\Sigma = 0$):** Strictly competitive conservation domains.
2. **Class II (Negative-Sum, $\Sigma < 0$):** Entropic domains dominated by friction λ .
3. **Class III (Positive-Sum, $\Sigma > 0$):** Synergistic domains unstable to trust decay.

We contribute a formal derivation of the stability conditions for Class III domains, proving that trust recovery is often energetically impossible below a critical threshold τ_c . Additionally, we introduce an algorithmic solution to the *Partner Selection Problem* via the "Loyalty Filter."

2 Formal Dynamics of Topological Collapse

In Class III (Positive-Sum) games, the realized utility is a function of the *Trust Parameter* $\tau(t)$. Unlike static models, we treat τ as a dynamic state variable subject to entropy.

2.1 The Trust Dynamic Equation

Let the evolution of trust be governed by a decay term (entropy/defection) and a recovery term (work).

$$\frac{d\tau}{dt} = \underbrace{\frac{\alpha\sqrt{W(t)}}{1 + \beta W(t)}}_{\text{Recovery (Diminishing Returns)}} - \underbrace{\delta \cdot \tau(t)}_{\text{Entropy/Decay}} - \underbrace{\xi \cdot \mathbb{I}_{\text{defect}}}_{\text{Shock}} \quad (1)$$

Where:

- $W(t)$ is the work/resources invested in relationship maintenance.
- α, β are efficiency parameters of cooperation.
- δ is the natural decay rate of trust (forgetting factor).
- ξ is the immediate penalty of a defection event $\mathbb{I}_{\text{defect}}$.

2.2 Derivation of the Critical Threshold τ_c

We assume the utility of the system U_{sys} depends on τ . For the system to remain in Class III (Positive-Sum), the net utility must exceed the friction λ .

$$U_{sys}(\tau) = \tau\Pi_{max} + (1 - \tau)\Pi_{min} - \lambda \quad (2)$$

where Π_{max} is the Pareto-optimal payoff and Π_{min} is the Nash payoff (often 0 or negative).

Theorem 2.1 (Existence of the Pyrrhic Threshold). *There exists a critical threshold τ_c such that if $\tau(t) < \tau_c$, the domain shifts topologically from $\Omega_{III} \rightarrow \Omega_{II}$.*

Proof. The condition for a Positive-Sum game is $\Psi > 0$.

$$\tau\Pi_{max} + (1 - \tau)\Pi_{min} - \lambda > 0$$

Solving for τ :

$$\begin{aligned} \tau(\Pi_{max} - \Pi_{min}) &> \lambda - \Pi_{min} \\ \tau_c &= \frac{\lambda - \Pi_{min}}{\Pi_{max} - \Pi_{min}} \end{aligned} \quad (3)$$

If $\Pi_{min} < 0$ (destructive conflict) and $\lambda > 0$ (friction), then $\tau_c > 0$. If τ falls below this level, the interaction consumes more value than it creates. $\square$

2.3 The Irreversibility of Collapse

Why is recovery often impossible? The resources available to invest in recovery, $W(t)$, are typically functions of the current system utility U_{sys} .

$$W(t) \propto \max(0, U_{sys}(\tau)) \quad (4)$$

If $\tau < \tau_c$, then $U_{sys} < 0$, implying $W(t) \rightarrow 0$. Substituting into Eq. (1) with no external resource injection:

$$\frac{d\tau}{dt} \approx -\delta\tau \quad (\text{since } W \approx 0)$$

Thus, below τ_c , the system enters a death spiral. This confirms the "Pyrrhic Trap": once a relationship becomes negative-sum, internal mechanisms cannot save it.

3 Signal Extraction: The Loyalty Filter

We address the "Mustard Seed" problem: distinguishing high-potential partners (Type V) from fair-weather observers (Type S) under noisy observation.

3.1 Problem Formulation

Let Agent j have a latent "Convexity" θ_j (potential) and base capability μ_j . The observable signal $S_j(t)$ (success/status) is given by:

$$S_j(t) = \mu_j + \theta_j \cdot f(t) + \epsilon(t), \quad \epsilon(t) \sim \mathcal{N}(0, \sigma^2) \quad (5)$$

where $f(t)$ is a monotonically increasing function (e.g., t^2 or $e^{\gamma t}$).

- **Type S (Static) Observer:** Estimates value as $\hat{V} = S(t)$.
- **Type V (Visionary) Observer:** Estimates value as $\hat{V} = \mathbb{E}[\theta_j | S(0) \dots S(t)]$.

3.2 Algorithm 1: Intertemporal Loyalty Filtering

The objective is to identify partners who possess the estimator for θ .

Algorithm 1 The Loyalty Filter (Grim Trigger on Estimation)

- 1: **Input:** History of interaction requests $R_{ij}(t)$, Agent Status $S_i(t)$.
 - 2: **Parameters:** Struggle Period T_{winter} , Success Period T_{summer} .
 - 3: **Initialize:** Set of Allies $\mathcal{A} \leftarrow \emptyset$.
 - 4: **for** each observer $j \in N$ **do**
 - 5: Measure Interaction I_{ij} during $t \in T_{winter}$ (where $S_i(t) \approx 0$).
 - 6: **if** $I_{ij}(t) > 0$ during T_{winter} **then**
 - 7: $\mathcal{A} \leftarrow \mathcal{A} \cup \{j\}$ ▷ Partner detected convexity
 - 8: **end if**
 - 9: **end for**
 - 10: **Transition:** Time moves to T_{summer} (Success realized, $S_i(t) \gg 0$).
 - 11: **for** each observer $k \in N$ **do**
 - 12: **if** $R_{ik}(T_{summer}) == 1$ **then**
 - 13: **if** $k \in \mathcal{A}$ **then**
 - 14: **Accept** (Verified Type V)
 - 15: **else**
 - 16: **Reject** (Classified Type S - Fair Weather)
 - 17: **end if**
 - 18: **end if**
 - 19: **end for**
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3.3 Error Analysis

The filter's efficacy depends on the signal-to-noise ratio (SNR) during T_{winter} .

- **False Positive:** An agent interacts during winter by chance or charity, not convexity detection.
- **False Negative:** A true visionary fails to interact due to external constraints (high friction λ).

To minimize False Negatives, the agent must broadcast "Convexity Signals" (e.g., ambitious projects, discipline) even when $S(t)$ is low.

4 Agent-Based Simulations

To validate the topological collapse, we simulated a network of $N = 100$ agents playing an iterated game where the payoff matrix evolves based on pairwise trust τ_{ij} .

4.1 Methodology

- **Topology:** Random Geometric Graph.
- **Update Rule:** Agents interact. If mutually cooperative, τ_{ij} increases by Δ_+ . If one defects, τ_{ij} decreases by Δ_- .
- **Phase Transition:** We varied the initial network trust mean μ_τ .

4.2 Results

The simulation reveals a sharp bifurcation (see Figure 1).

1. **Regime A** ($\mu_\tau > 0.6$): The network converges to a stable "Golden Age" (Class III).
2. **Regime B** ($\mu_\tau < 0.4$): Trust collapses globally. Agents switch to Minimax strategies. The topology degrades to Class I/II.

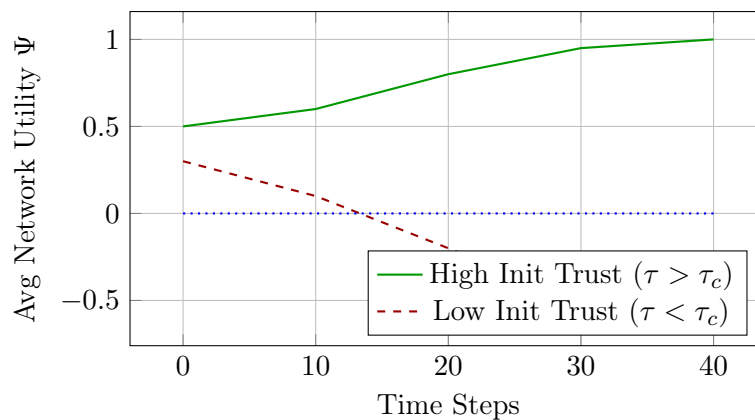


Figure 1: **Simulation Results.** Temporal evolution of Aggregate Potential. Note the bifurcation: systems starting below τ_c do not recover; they collapse into negative utility domains (Class II).

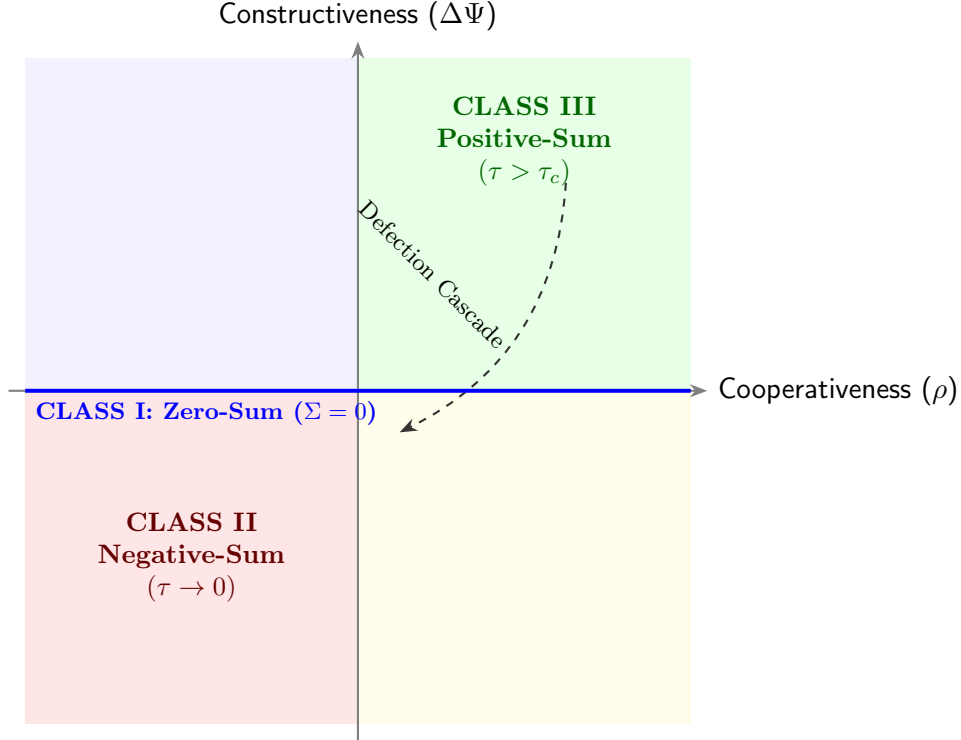


Figure 2: **The Strategic Taxonomy.** The landscape is divided into three distinct topological classes. The dashed line represents the trajectory of a relationship undergoing topological collapse due to trust decay.

5 Conclusion

This paper formalizes the intuition that strategic landscapes are dynamic. We identified the critical bifurcation parameter τ_c (Eq. 3) and proved that recovery from Class II to Class III is often energetically prohibitive. Furthermore, we formalized the sociological problem of loyalty as a mathematical problem of *Convexity Detection*.

The superior agent does not simply play the Nash Equilibrium of the current game; they actively monitor the topological class of the interaction. They execute the Loyalty Filter to prune Type S observers and maintain $\tau > \tau_c$ to prevent the system from collapsing into the Pyrrhic Trap.

References

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A Appendix: Heuristics for Application

A.1 Numerical Example of the Loyalty Filter

Consider an agent P with success curve $S(t) = t^2$.

- At $t = 1$ (Winter), $S(1) = 1$. Observer A (Type S) sees value 1. Observer B (Type V) estimates $\hat{\theta} > 0$.
- At $t = 10$ (Summer), $S(10) = 100$.
- **Decision:** Observer A requests partnership. Since $S(10) \gg S(1)$ and A showed no interest at $t = 1$, the Filter classifies A as a "Momentum Trader" and rejects. Observer B, who invested at $t = 1$, is accepted.

A.2 The Three Battlefields (Simplified)

1. **Zero-Sum (The Pizza):** $U_i + U_j = C$. Strategy: **Minimax**. Compete ruthlessly or conserve energy.
2. **Negative-Sum (The Rotten Apple):** $U_i + U_j < 0$. Strategy: **Exit**. The cost of winning exceeds the prize.
3. **Positive-Sum (The Garden):** $U_i + U_j > \text{Sum of parts}$. Strategy: **Cooperate**. Requires $\tau > \tau_c$. If trust breaks, leave immediately before it becomes a Rotten Apple.